LOWER SHRIEK FOR LOCALLY QUASI-FINITE FINITELY PRESENTED MORPHISMS PRESERVES CONSTRUCTIBILITY

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Construction of the lower shriek functor on derived categories for locally quasi-finite morphisms (Lemma lemma-lqf-shriek-derived, canonical tag 0F5N). cited at statement lines 4690-4691. The functor f_! whose constructibility preservation is claimed is the functor constructed there.
- Colimit description of lower shriek (Lemma lemma-lqf-colimit-f-shriek, canonical tag 0F5H). cited at proof lines 4704-4708. Reduces the claim to the pieces f_{i,!} and f_{ii',!} over the finite affine covering.
- Zariski main theorem factorization through a finite morphism (More on Morphisms, Lemma more-morphisms-lemma-quasi-finite-separated-pass-through-finite-addendum, canonical tag 0F2N). cited at proof lines 4711-4716. Supplies the factorization f = g after j with j an open immersion and g finite of finite presentation.
- Composition of lower shriek (Lemma lemma-f-shriek-composition, canonical tag 0F50). cited at proof lines 4716-4717. Gives f_! = g_! after j_! for the factorization.
- Constructibility of lower shriek along open immersions (Etale Cohomology, Lemma etale-cohomology-lemma-jshriek-constructible, canonical tag 03S8). cited at proof lines 4718-4719. j_! of the constructible sheaf F is constructible on X'.
- Agreement of lower shriek with pushforward for proper morphisms (Lemma lemma-proper-f-shriek, canonical tag 0F51). cited at proof lines 4720-4721. The finite morphism g satisfies g_! = g_*.
- Constructibility of pushforward along finite morphisms (Etale Cohomology, Lemma etale-cohomology-lemma-finite-pushforward-constructible, canonical tag 095R). cited at proof lines 4722-4725. g_* of the constructible sheaf is constructible, which closes the argument.

Source reference. Original lemma 0F5N: sources/stacks-chow/tags.txt, source lines 18943–18943. Cited at statement lines 4690-4691: the functor f_! whose constructibility preservation is claimed is the functor constructed there.

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Source reference. Original lemma 0F5H: sources/stacks-chow/tags.txt, source lines 18937–18937. Cited at proof lines 4704-4708: reduces the claim to the pieces f_{i,!} and f_{ii',!} attached to the finite affine covering.

Source reference. Original lemma 0F50: sources/stacks-chow/tags.txt, source lines 18920–18920. Cited at proof lines 4716-4717: gives f_! = g_! \circ j_! for the factorization f = g \circ j.

Source reference. Original lemma 0F51: sources/stacks-chow/tags.txt, source lines 18921–18921. Cited at proof lines 4720-4721: the finite morphism g satisfies g_! = g_*.

Source reference. Original lemma 0F2N: sources/stacks-chow/tags.txt, source lines 18838–18838. Cited at proof lines 4711-4716: supplies the factorization f = g \circ j with j an open immersion and g finite of finite presentation.

Source reference. Original definition 01TP: sources/stacks-chow/tags.txt, source lines 2382–2382. Cited at proof lines 4701-4702: X is quasi-compact and quasi-separated when f is locally quasi-finite and of finite presentation.

Source reference. Original lemma 03S8: sources/stacks-chow/tags.txt, source lines 4957–4957. Cited at proof lines 4718-4719: j_!F is constructible on X'.

Source reference. Original lemma 095R: sources/stacks-chow/tags.txt, source lines 11594–11594. Cited at proof lines 4722-4725: g_*j_!F is constructible, which closes the argument.

Lemma 0.2. *Let $f : X \to Y$ be a morphism of schemes which is locally quasi-finite and of finite presentation. The functor $f_! : D(X_{étale}, \Lambda) \to D(Y_{étale}, \Lambda)$ of Lemma 0F5N sends $D_c(X_{étale}, \Lambda)$ into $D_c(Y_{étale}, \Lambda)$.*

Proof. Since the functor $f_!$ is exact, it suffices to show that $f_!\mathcal{F}$ is constructible for any constructible sheaf $\mathcal{F}$ of Λ-modules on $X_{étale}$. The question is local on Y and hence we may and do assume Y is affine. Then X is quasi-compact and quasi-separated, see Morphisms, Definition 01TP. Say $X = \bigcup_{i=1,\ldots,n} X_i$ is a finite affine open covering. By Lemma 0F5H we see that it suffices to show that $f_{i,!}\mathcal{F}|_{X_i}$ and $f_{ii',!}\mathcal{F}|_{X_i \cap X_{i'}}$ are constructible where $f_i : X_i \to Y$ and $f_{ii'} : X_i \cap X_{i'} \to Y$ are the restrictions of f. Since X_i and $X_i \cap X_{i'}$ are quasi-compact and separated this means we may assume f is separated. By Zariski's main theorem (in the form of More on Morphisms, Lemma 0F2N) we can choose a factorization $f = g \circ j$ where $j : X \to X'$ is an open immersion and $g : X' \to Y$ is finite and of finite presentation. Then $f_! = g_! \circ j_!$ by Lemma 0F50. By Étale Cohomology, Lemma 03S8 we see that $j_!\mathcal{F}$ is constructible on X'. The morphism g is finite hence $g_! = g_*$ by Lemma 0F51. Thus $f_!\mathcal{F} = g_!j_!\mathcal{F} = g_*j_!\mathcal{F}$ is constructible by Étale Cohomology, Lemma 095R. □